Построение операционного слоя данных (Silver Layer)

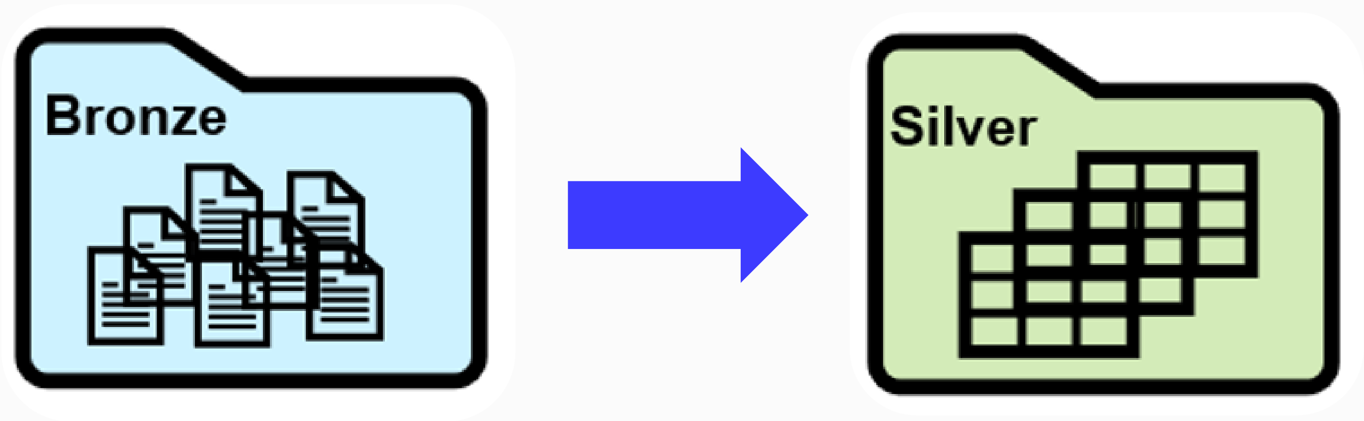
Skillbox

Цели

1. Узнать про нюансы построения слоя операционных данных:
   - валидация по схеме
   - выделение сущностей
   - нейминг
   - очистка
   - регистрация в каталоге
2. Научиться наполнять слой сырых данных

Валидация по схеме

Данные csv
Чикаго криминал



	<anonymous>	ID	Case Number	Date	Block	IUCR	Primary Type	Desc
1	388	4785	HP610824	10/07/2008 12:39:00 PM	000XX E 75TH ST	0110	HOMICIDE	FIRS
2	835	4786	HP616595	10/09/2008 03:30:00 AM	048XX W POLK ST	0110	HOMICIDE	FIRS
3	1334	4787	HP616904	10/09/2008 08:35:00 AM	030XX W MANN DR	0110	HOMICIDE	FIRS
4	1907	4788	HP618616	10/10/2008 02:33:00 AM	052XX W CHICAGO AVE	0110	HOMICIDE	FIRS
5	2436	4789	HP619020	10/10/2008 12:50:00 PM	026XX S HOMAN AVE	0110	HOMICIDE	FIRS
6	3056	4790	HP620131	10/10/2008 08:32:00 PM	015XX W 14TH ST	0110	HOMICIDE	FIRS
7	3611	4791	HP620406	10/11/2008 12:55:00 AM	020XX W 23RD ST	0110	HOMICIDE	FIRS
8	4425	4792	HP622040	10/11/2008 10:25:00 PM	069XX S PAXTON AVE	0110	HOMICIDE	FIRS
9	5213	4793	HP622164	10/11/2008 10:00:00 PM	060XX S SAWYER AVE	0110	HOMICIDE	FIRS
10	5867	4794	HP622189	10/12/2008 05:47:00 AM	024XX W 51ST ST	0110	HOMICIDE	FIRS
11	6359	4795	HP623937	10/12/2008 10:33:00 PM	057XX S ASHLAND AVE	0110	HOMICIDE	FIRS
12	6917	4796	HP624032	10/12/2008 10:55:00 PM	004XX E 79TH ST	0110	HOMICIDE	FIRS
13	7483	4797	HP625052	10/13/2008 02:49:00 PM	112XX S VERNON AVE	0110	HOMICIDE	FIRS
14	7955	4798	HP622688	10/13/2008 07:04:00 PM	024XX S HAMLIN AVE	0110	HOMICIDE	FIRS

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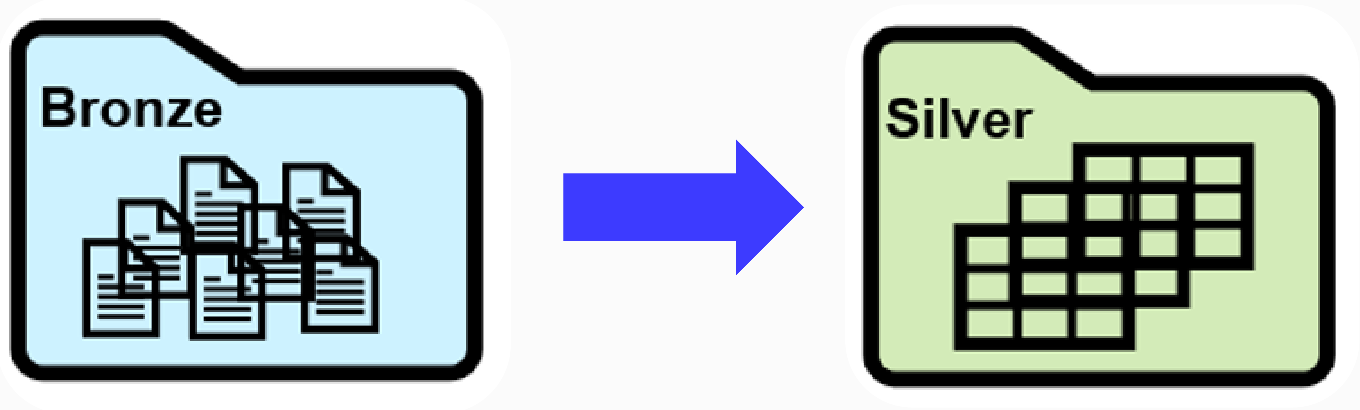
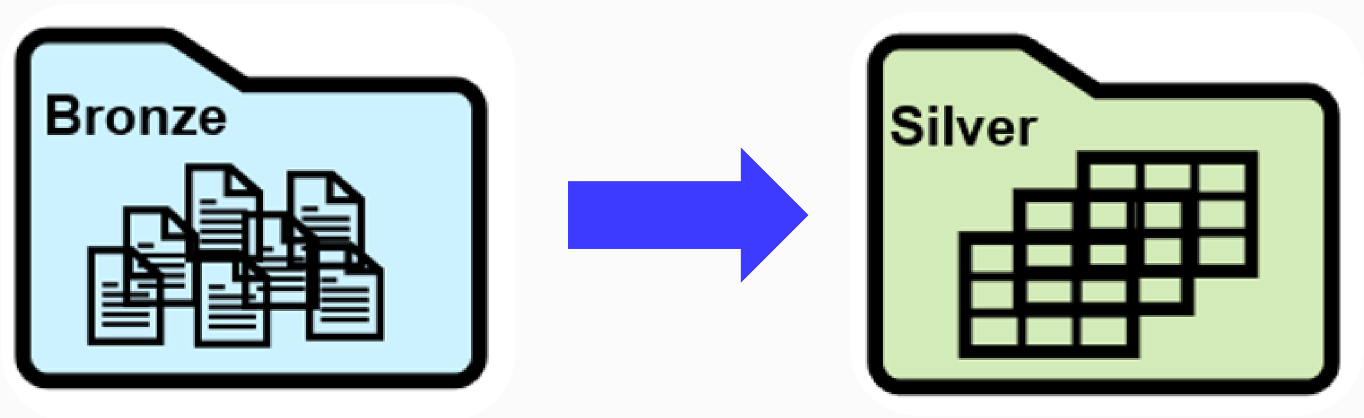
Схема вручную




```
schema = StructType([
    StructField("ID", IntegerType(), nullable=False),
    StructField("Case Number", StringType(), nullable=False),
    StructField("Date", StringType(), nullable=True),
    StructField("Block", StringType(), nullable=True),
    StructField("IUCR", IntegerType(), nullable=True),
    StructField("Primary Type", StringType(), nullable=True),
    StructField("Description", StringType(), nullable=True),
    StructField("Location Description", StringType(), nullable=True),
    StructField("Arrest", BooleanType(), nullable=True)
])
```

Изображения: Skillbox

Схема вручную



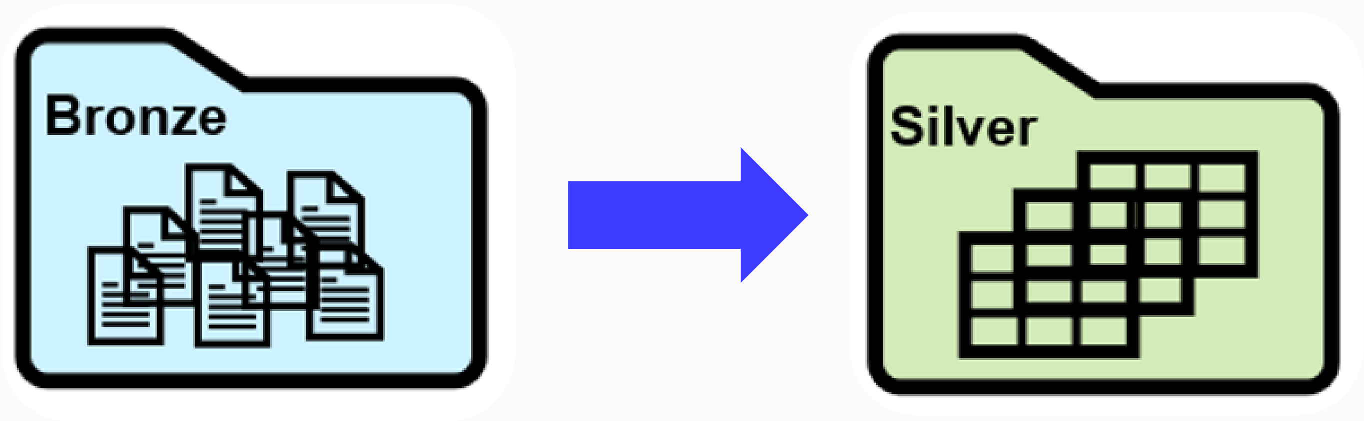

```
silver_manual_df = (spark
                .read
                .format("csv")
                .schema(schema)
                .option("header", "true")
                .load("../sources/data_lake/source2"))


silver_manual_df.printSchema()
```

```
root
 |-- ID: integer (nullable = true)
 |-- Case Number: string (nullable = true)
 |-- Date: string (nullable = true)
 |-- Block: string (nullable = true)
 |-- IUCR: integer (nullable = true)
```

Изображения: Skillbox

Схема автоматически




```
silver_df = spark.read.
    format("csv").
    option("inferSchema", "true").
    option("header", "true").
    load("../sources/data_lake/source2")

silver_df.printSchema()
```

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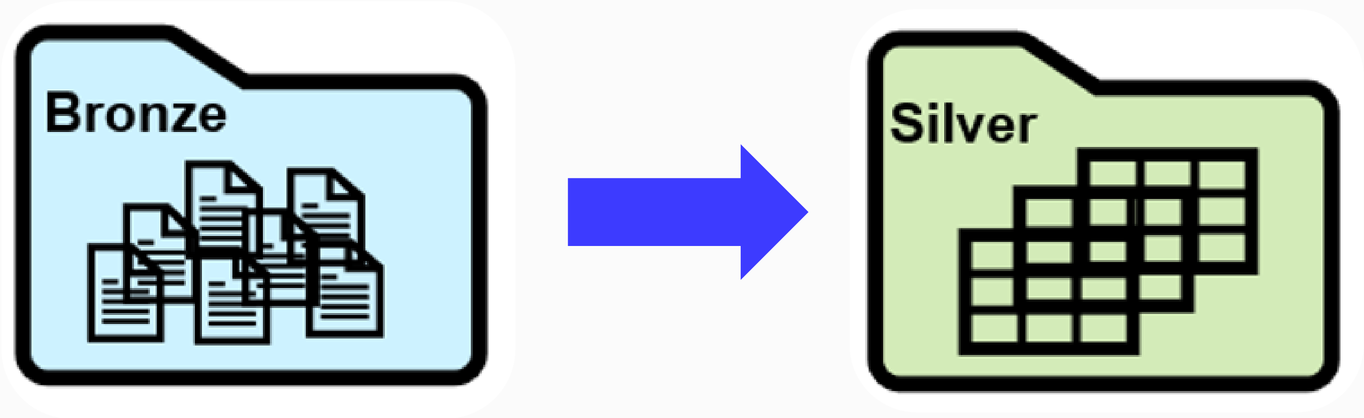
Схема автоматически

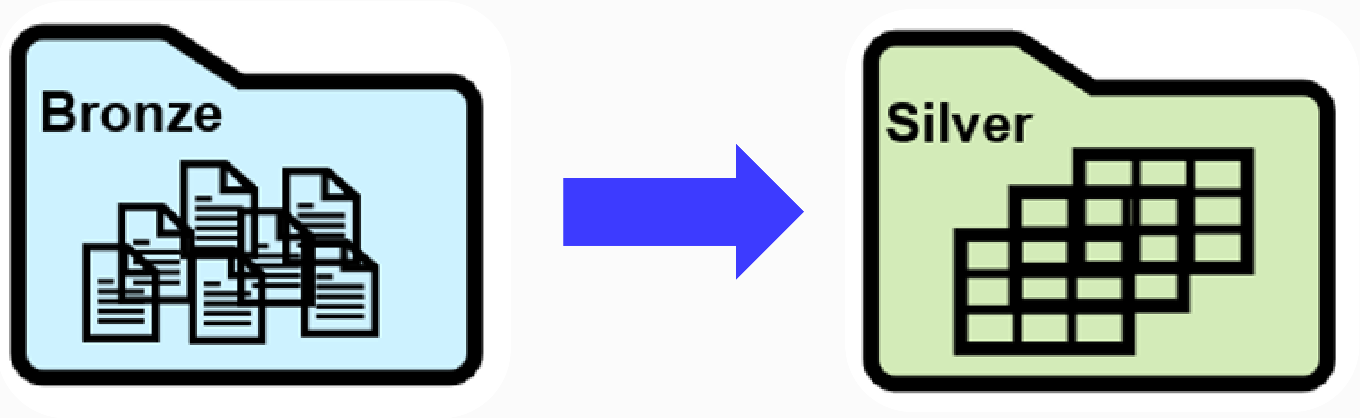



```
root
 |-- ID: integer (nullable = true)
 |-- Case Number: string (nullable = true)
 |-- Date: string (nullable = true)
 |-- Block: string (nullable = true)
 |-- IUCR: integer (nullable = true)
  .....................................
 |-- Latitude: double (nullable = true)
 |-- Longitude: double (nullable = true)
 |-- Location: string (nullable = true)
```

Изображения: Skillbox

Выделение сущности

Выделение колонок



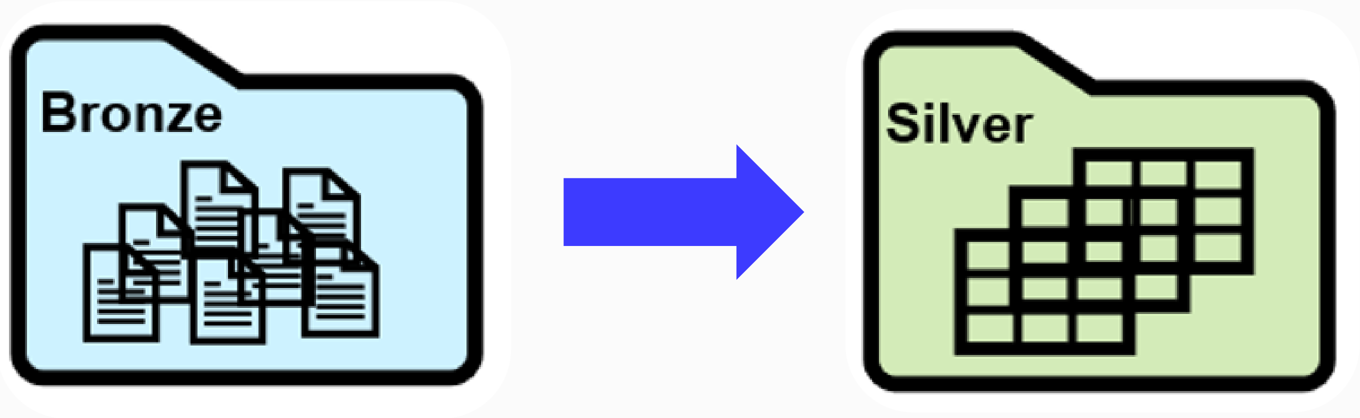

```
arrested_df = silver_df
        .select("ID", "Case Number", "Date", "Arrest")

arrested_df.printSchema()
```

```
root
 |-- ID: integer (nullable = true)
 |-- Case Number: string (nullable = true)
 |-- Date: string (nullable = true)
 |-- Arrest: boolean (nullable = true)
```

Изображения: Skillbox

Порядок колонок



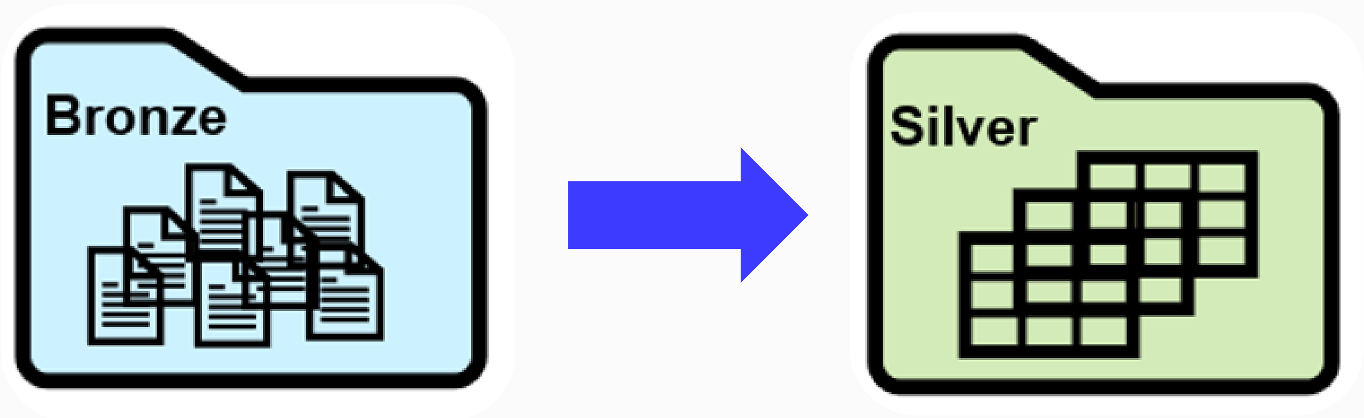

```
arrested_df = silver_df
        .select("Date", "ID", "Case Number", "Arrest")

arrested_df.printSchema()
```

```
root
 |-- Date: string (nullable = true)
 |-- ID: integer (nullable = true)
 |-- Case Number: string (nullable = true)
 |-- Arrest: boolean (nullable = true)
```

Изображения: Skillbox

Выделение колонок




```
arrested_df = silver_df
        .select("ID",
                        "Case Number",
                        "Date",
                        "Arrest")

arrested_df.show()
```

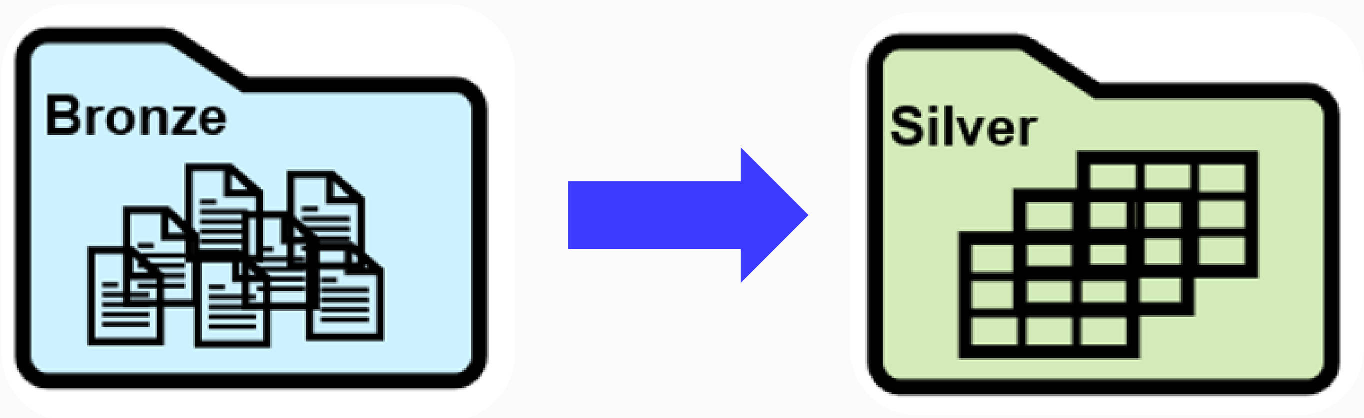
```
+----+-----------+--------------------+------+
|  ID|Case Number|                Date|Arrest|
+----+-----------+--------------------+------+
|4785|   HP610824|10/07/2008 12:39:...|  true|
|4785|   HP610824|10/07/2008 12:39:...|  true|
|4786|   HP616595|10/09/2008 03:30:...|  true|
|4787|   HP616904|10/09/2008 08:35:...| false|
|4788|   HP618616|10/10/2008 02:33:...| false|
|4789|   HP619020|10/10/2008 12:50:...| false|
|4790|   HP620131|10/10/2008 08:32:...|  true|
|4791|   HP620406|10/11/2008 12:55:...|  true|
|4792|   HP622040|10/11/2008 10:25:...| false|
|4793|   HP622164|10/11/2008 10:00:...| false|
|4794|   HP622189|10/12/2008 05:47:...| false|
|4795|   HP623937|10/12/2008 10:33:...| false|
```

Изображения: Skillbox

Фильтрация

```
arrested_df = silver_df
        .select("ID",
                    "Case Number",
                    "Date",
                    "Arrest")
.where("Arrest = true")

arrested_df.show()
```




```
+----+-----------+--------------------+------+
|  ID|Case Number|                Date|Arrest|
+----+-----------+--------------------+------+
|4785|   HP610824|10/07/2008 12:39:...|  true|
|4785|   HP610824|10/07/2008 12:39:...|  true|
|4786|   HP616595|10/09/2008 03:30:...|  true|
|4790|   HP620131|10/10/2008 08:32:...|  true|
|4791|   HP620406|10/11/2008 12:55:...|  true|
|4799|   HP626379|10/14/2008 09:37:...|  true|
|4800|   HP626784|10/14/2008 01:27:...|  true|
|4801|   HP628839|10/15/2008 04:10:...|  true|
|4802|   HP631705|10/16/2008 09:35:...|  true|
|4803|   HP626520|10/16/2008 04:25:...|  true|
|4804|   HP631900|10/17/2008 05:17:...|  true|
```

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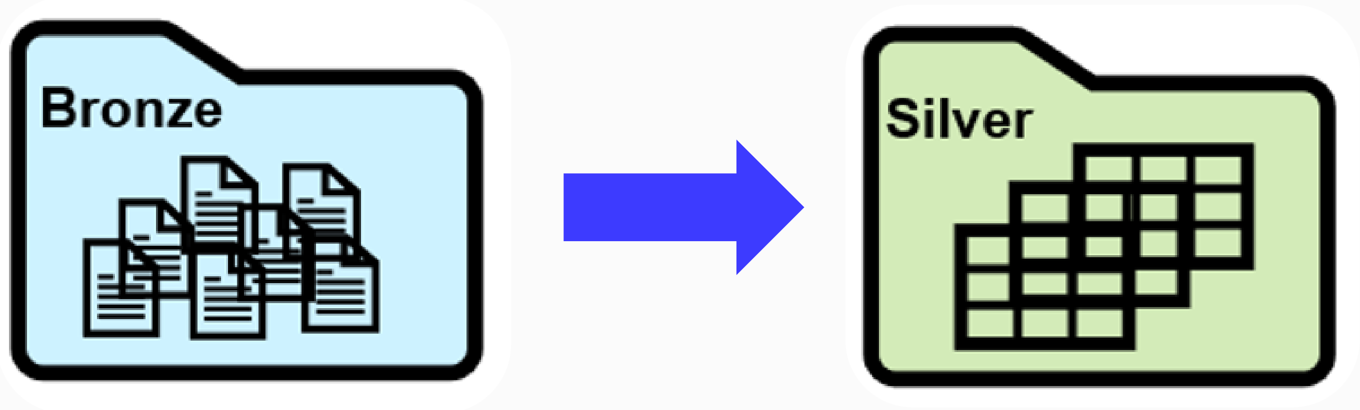
Renaming & Casting

Переименование колонки

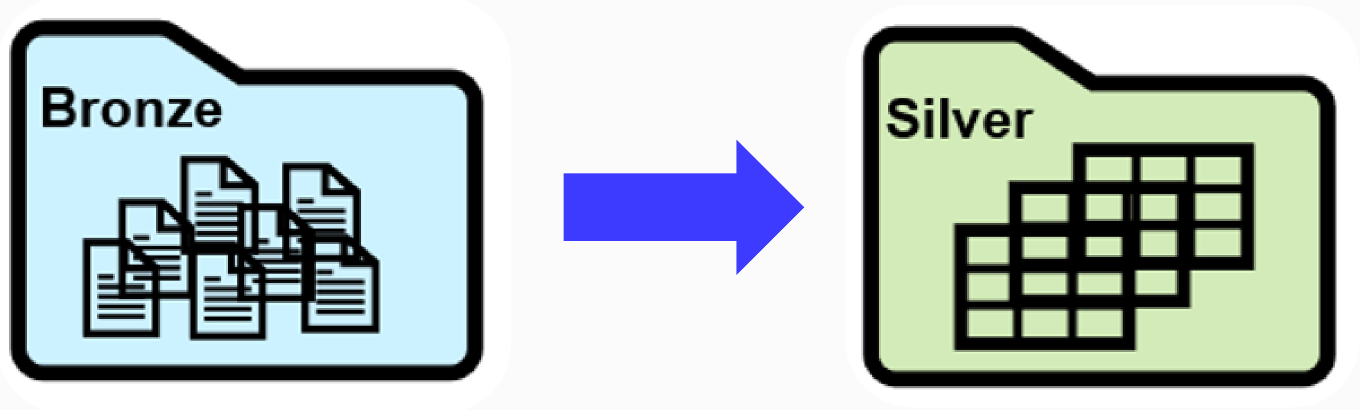



```
arrested_df = (silver_df.select(col("ID").alias("arrest_id"),
                                col("Case Number"),
                                col("Date"),
                                col("Arrest"))
               .where("Arrest = true"))
```

```
root
 |-- arrest_id: integer (nullable = true)
 |-- Case Number: string (nullable = true)
 |-- Date: string (nullable = true)
 |-- Arrest: boolean (nullable = true)
```

Изображения: Skillbox

Переименование колонки

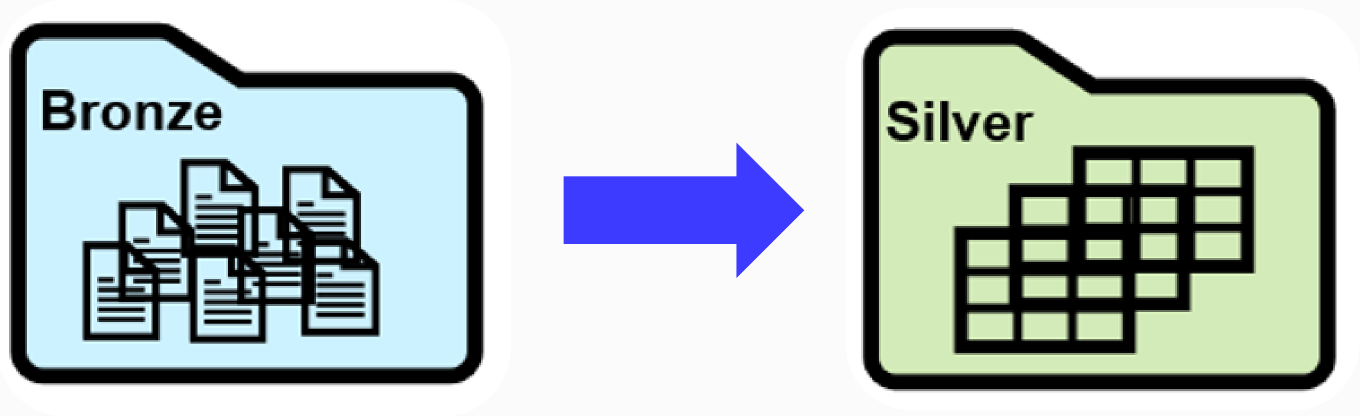



```
arrested_df = (silver_df.select(col("ID").alias("arrest_id"),
                                col("Case Number"),
                                col("Date"),
                                col("Arrest"))
      .withColumnRenamed("Case Number", "arrest_case_number")
            .where("Arrest = true"))
```

```
root
 |-- arrest_id: integer (nullable = true)
 |-- arrest_case_number: string (nullable = true)
 |-- Date: string (nullable = true)
 |-- Arrest: boolean (nullable = true)
```

Изображения: Skillbox

Добавление колонки

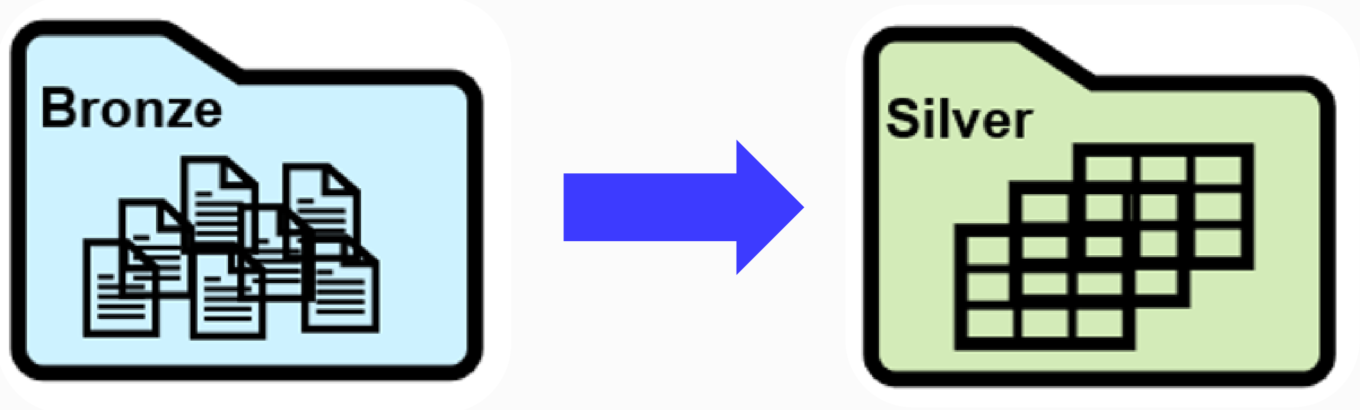



```
arrested_df = (silver_df.select(col("ID").alias("arrest_id"),
                                col("Case Number"),
                                col("Date"),
                                col("Arrest").)
        .withColumnRenamed("Case Number", "arrest_case_number")
        .withColumn("Date", "date")
        .where("Arrest = true"))
```

```
root
 |-- arrest_id: integer (nullable = true)
 |-- arrest_case_number: string (nullable = true)
 |-- date: string (nullable = true)
 |-- Arrest: boolean (nullable = true)
```

Изображения: Skillbox

Изменение типа колонки



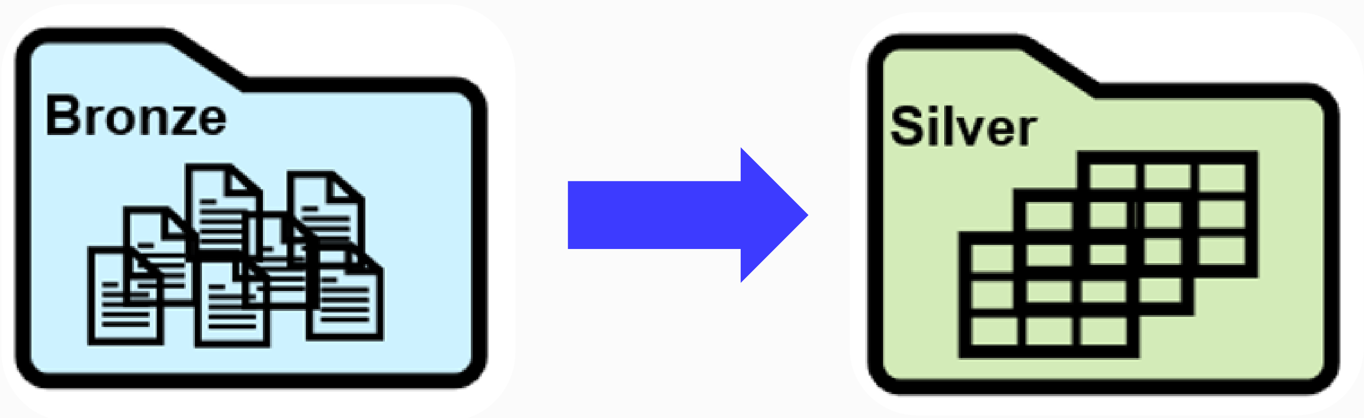

```
arrested_df = (silver_df.select(col("ID").alias("arrest_id").cast("int"),
                                col("Case Number"),
                                col("Date"),
                                col("Arrest"))
        .withColumnRenamed("Case Number", "arrest_case_number")
        .withColumn("Date", "date")
        .where("Arrest = true"))
```

```
root
 |-- ID: string (nullable = true)
 |-- arrest_case_number: string (nullable = true)
 |-- date: string (nullable = true)
 |-- Arrest: boolean (nullable = true)
```

Изображения: Skillbox

Очистка от дубликатов

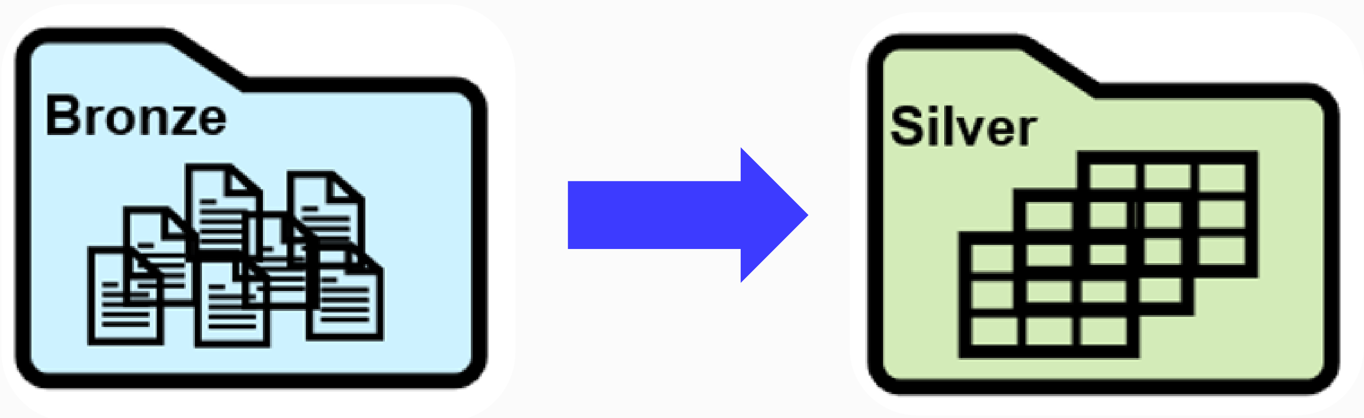
Полные дубли




```
arrested_df = (silver_df.select(col("ID").alias("arrest_id"),
                                                      col("Case Number"),
                                                      col("Date"),
                                                      col("Arrest"))
          .withColumnRenamed("Case Number", "arrest_case_number")
          .withColumn("Date", "date")
          .where("Arrest = true")
                     .distinct()
                         )
```

Изображения: Skillbox

По составному ключу

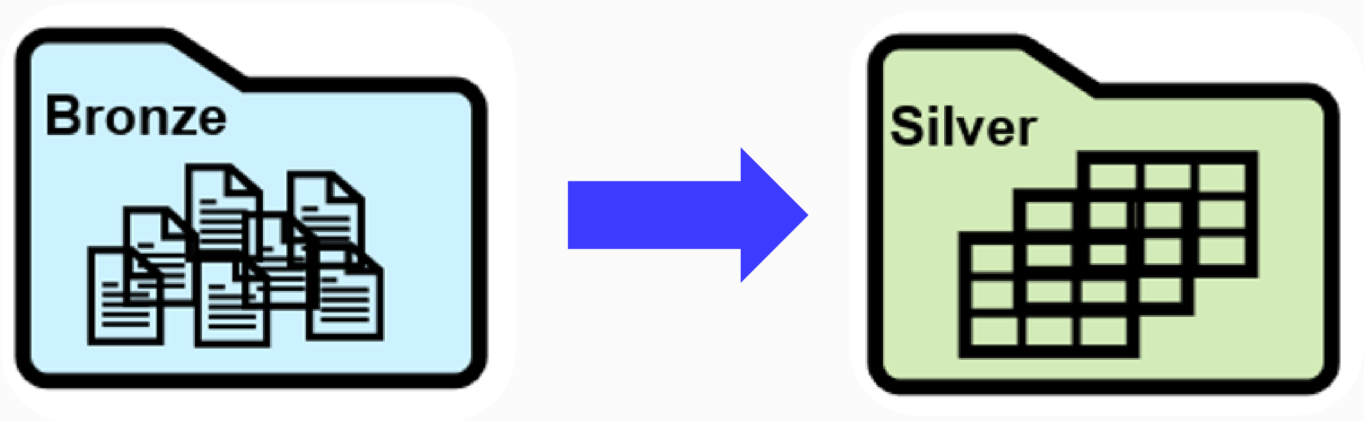



```
arrested_df = (silver_df.select(col("ID").alias("arrest_id"),
                                                   col("Case Number"),
                                                   col("Date"),
                                                   col("Arrest"))
          .withColumnRenamed("Case Number", "arrest_case_number")
          .withColumn("Date", "date")
          .where("Arrest = true")
                         .drop_duplicates("ID", "Arrest")
                         )
```

Изображения: Skillbox

Сохранение и регистрация в каталоге

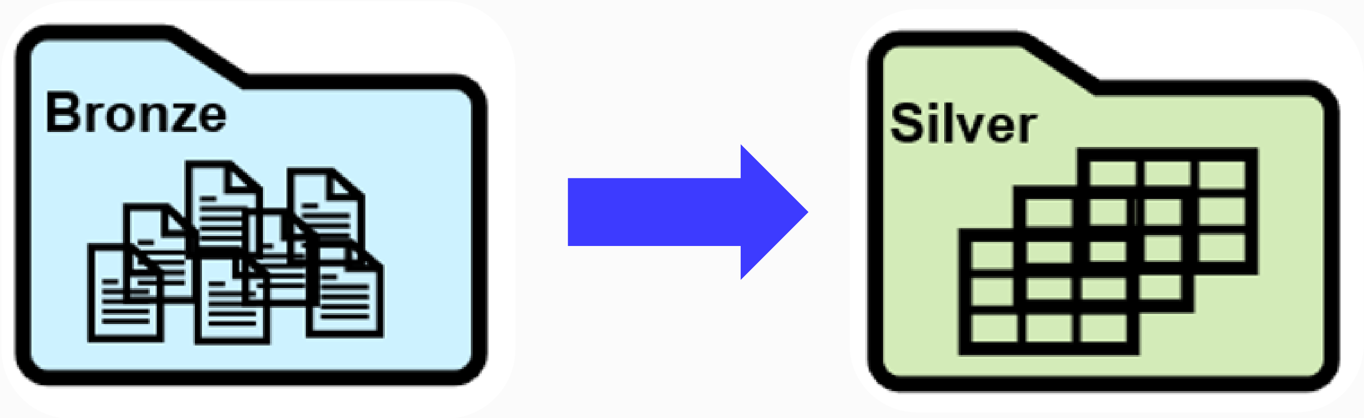
ORC




```
arrested_df
 .coalesce(1)
 .write
 .orc("../sources/data_lake/proc_layer")
```

Изображения: Skillbox

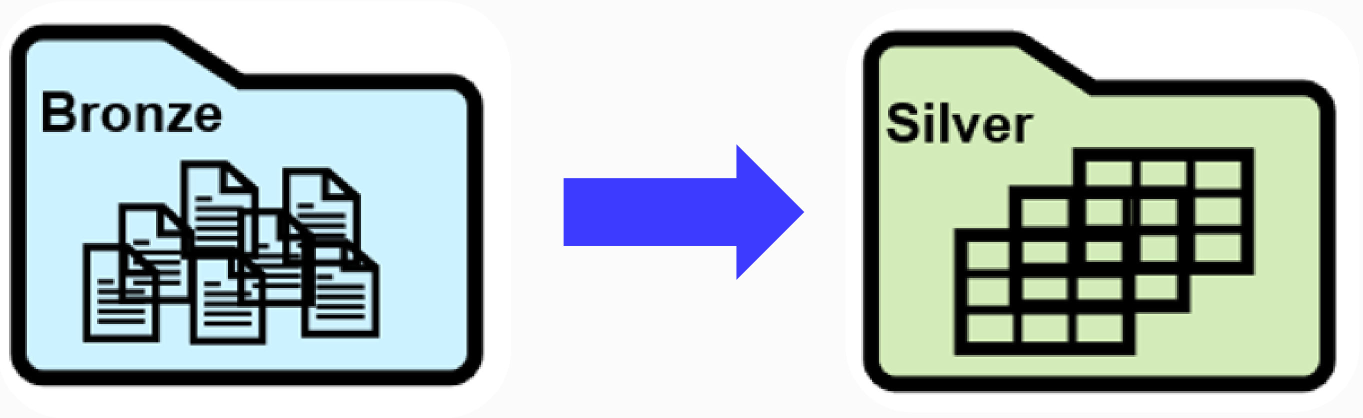
Parquet




```
arrested_df
 .coalesce(1)
 .write
 .parquet("../sources/data_lake/proc_layer"))
```

Изображения: Skillbox

Регистрация в каталоге




```
arrested_df
 .coalesce(1)
 .write
 .saveAsTable("proc_layer")
```

Изображения: Skillbox

Итоги

1. Узнали про нюансы построения слоя операционных данных:
   - валидация по схеме
   - выделение сущностей
   - нейминг
   - очистка
   - регистрация в каталоге
2. Научились наполнять слой сырых данных